

Library Management System – Use Case Diagram

AIM

To draw the Use Case Diagram for the Library Management System using UML.

Description

This use case diagram represents interaction between Member and Admin with the Library Management System. It shows how users access functionalities like searching, issuing, and managing books. The system boundary defines the scope and include relationships ensure login is mandatory for key operations.

Documentation:

- a. Member registers and logs into the system.
- b. Member searches for books and views details.
- c. Member issues or returns books after login.
- d. Admin manages books and member records.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Use Case Diagram successfully represents the required aspect of the Library Management System.

Library Management System – Activity Diagram

AIM

To draw the Activity Diagram for the Library Management System using UML.

Description

This diagram represents workflow of operations from login to book issue or return. It shows sequence of activities and decision points such as book availability and user actions.

Documentation:

- a. User logs into the system.
- b. User searches for a book.
- c. System checks availability.
- d. Book is issued or returned and database is updated.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Activity Diagram successfully represents the required aspect of the Library Management System.

Library Management System – State Chart Diagram

AIM

To draw the State Chart Diagram for the Library Management System using UML.

Description

This diagram represents different states of a book during its lifecycle. It shows how the book transitions between states based on events like issue and return.

Documentation:

- a. Book is initially in Available state.
- b. Book moves to Issued state when borrowed.
- c. Book enters Overdue state if not returned on time.
- d. Book returns to Available state after return.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The State Chart Diagram successfully represents the required aspect of the Library Management System.

Library Management System – Class Diagram

AIM

To draw the Class Diagram for the Library Management System using UML.

Description

This diagram represents static structure of the system with classes like Member, Book, Transaction and Admin along with their attributes and methods.

Documentation:

- a. Member class stores user details.
- b. Book class maintains book information.
- c. Transaction handles issue and return.
- d. Admin manages system records.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Class Diagram successfully represents the required aspect of the Library Management System.

Library Management System – Sequence Diagram

AIM

To draw the Sequence Diagram for the Library Management System using UML.

Description

This diagram shows interaction between Member, System and Database in a time sequence for operations like issuing a book.

Documentation:

- a. Member sends request to system.
- b. System verifies request.
- c. Database is accessed for validation.
- d. Response is sent back to member.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Sequence Diagram successfully represents the required aspect of the Library Management System.

Library Management System – Component Diagram

AIM

To draw the Component Diagram for the Library Management System using UML.

Description

This diagram represents system modules like UI, Authentication, Book Management and Database and their dependencies.

Documentation:

- a. User interacts with UI component.
- b. Authentication verifies user login.
- c. Book module processes operations.
- d. Database stores and retrieves data.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Component Diagram successfully represents the required aspect of the Library Management System.

Library Management System – Deployment Diagram

AIM

To draw the Deployment Diagram for the Library Management System using UML.

Description

This diagram represents physical architecture including client device, server and database.

Documentation:

- a. Client sends request to server.
- b. Server processes application logic.
- c. Database stores system data.
- d. Response is returned to client.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Deployment Diagram successfully represents the required aspect of the Library Management System.

Passport Automation System – Use Case Diagram

AIM

To draw the Use Case Diagram for the Passport Automation System using UML.

Description

This diagram represents interaction between Applicant, Admin and Police in passport processing system.

Documentation:

- a. Applicant submits application.
- b. Admin processes request.
- c. Police verifies details.
- d. Passport is approved or rejected.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Use Case Diagram successfully represents the required aspect of the Passport Automation System.

Passport Automation System – Activity Diagram

AIM

To draw the Activity Diagram for the Passport Automation System using UML.

Description

This diagram shows workflow of passport processing from application to approval.

Documentation:

- a. Applicant fills application form.
- b. System verifies details.
- c. Police verification is done.
- d. Passport is approved or rejected.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Activity Diagram successfully represents the required aspect of the Passport Automation System.

Passport Automation System – State Chart Diagram

AIM

To draw the State Chart Diagram for the Passport Automation System using UML.

Description

This diagram shows states of passport application during processing.

Documentation:

- a. Application starts in Applied state.
- b. Moves to Verification state.
- c. Goes to Approved or Rejected state.
- d. Final status is updated.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The State Chart Diagram successfully represents the required aspect of the Passport Automation System.

Passport Automation System – Class Diagram

AIM

To draw the Class Diagram for the Passport Automation System using UML.

Description

This diagram represents classes like Applicant, Passport and Verification.

Documentation:

- a. Applicant stores personal details.
- b. Passport stores passport data.
- c. Verification checks records.
- d. Admin manages approvals.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Class Diagram successfully represents the required aspect of the Passport Automation System.

Passport Automation System – Sequence Diagram

AIM

To draw the Sequence Diagram for the Passport Automation System using UML.

Description

This diagram shows interaction between Applicant, System, Admin and Police.

Documentation:

- a. Applicant sends request.
- b. System processes data.
- c. Admin reviews application.
- d. Police verifies and responds.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Sequence Diagram successfully represents the required aspect of the Passport Automation System.

Passport Automation System – Component Diagram

AIM

To draw the Component Diagram for the Passport Automation System using UML.

Description

This diagram shows modules like UI, Processing, Verification and Database.

Documentation:

- a. UI collects user data.
- b. Processing module handles logic.
- c. Verification module checks details.
- d. Database stores records.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Component Diagram successfully represents the required aspect of the Passport Automation System.

Passport Automation System – Deployment Diagram

AIM

To draw the Deployment Diagram for the Passport Automation System using UML.

Description

This diagram represents deployment of system on client, server and database.

Documentation:

- a. Client accesses system.
- b. Server processes requests.
- c. Database stores application data.
- d. System returns results to user.

Diagram

Draw the diagram in ArgoUML as per the above description.

Conclusion

The Deployment Diagram successfully represents the required aspect of the Passport Automation System.